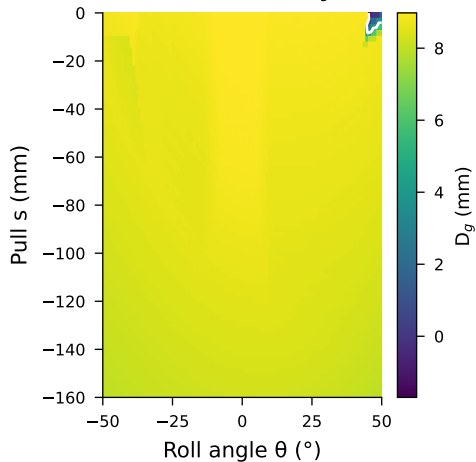
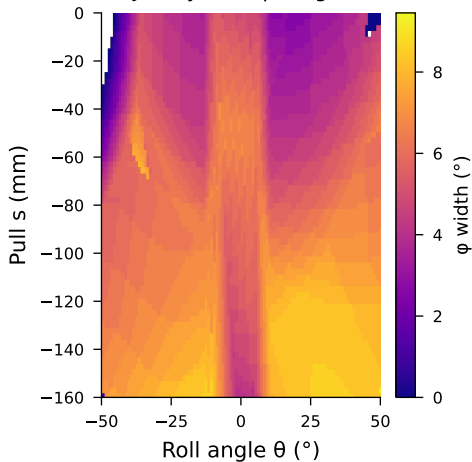


a Clearance atlas $D_g(\theta, s)$ b Jointly-safe ϕ range ($^{\circ}$)c Self-motion arc $d(\phi)$ 